BRAC University

Department of Computer Science and Engineering

Midterm Exam

Full Marks: **15 x 3 = 45**

Time: **1 hour 30 minutes**

Date: **27th July 2022**

Semester: **Summer 2022**

Course Code: **CSE460**

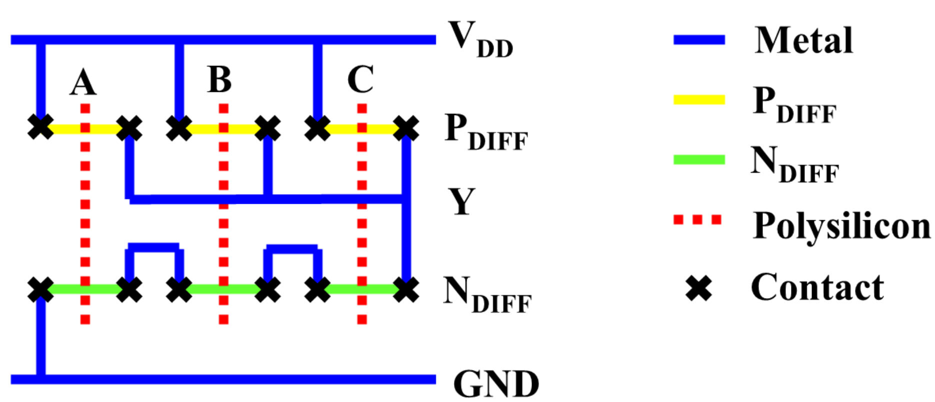
Course Title: **VLSI Design**

Set A

Student ID:	Name:	Section:

[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO2)	**(a)**	**Sketch** the transistor level schematic of the boolean logic function $F = \overline{A \cdot (B + C)}$	**3**
	(b)	From the schematic in part-a, **design** a stick diagram for the given logic function.	**7**
	(c)	A designer has created the stick diagram for a 3-input NAND gate. After a close inspection you see that the design is *not* area efficient and can be *further optimized*:  **Draw** the *optimized* stick diagram for this circuit.	**5**
2. (CO1)	In a CMOS fabrication process, two n-diffusion regions are required on the wafer. A VLSI engineer has designed a mask arrangement [figure 1(i)] to fabricate the desired pattern on a blank wafer [figure 1(ii)]. After the successful completion of photolithography and doping procedures, the desired structure was fabricated.		

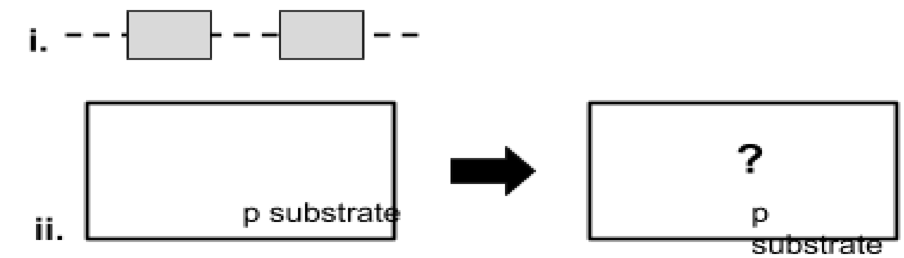
<table>
<tr><td></td><td colspan="2">

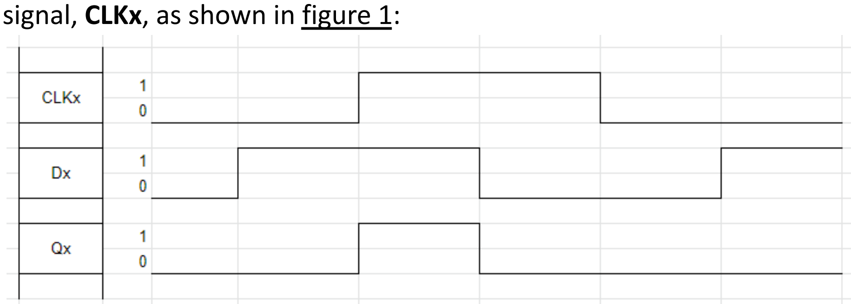
Figure 1: (i) n diffusion mask (top view)
(ii) Blank wafer (Cross-section view).
Figure 2: Cross-section view of the wafer after forming n diffusion regions.</td><td></td></tr>
<tr><td></td><td>(a)</td><td>Draw and complete figure 2. Properly label the different regions in the figure.</td><td>6</td></tr>
<tr><td></td><td>(b)</td><td>Briefly explain the following terms for the CMOS photolithography process:
(i) Photoresist
(ii) Etching (Stripping SiO2)</td><td>6</td></tr>
<tr><td></td><td>(c)</td><td>Why do we need contacts?</td><td>3</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td>3. (CO2)</td><td>(a)</td><td>“X” is a circuit element that produces output Qx from input Dx with a clock signal, CLKx, as shown in <u>figure 1</u>:


Figure 1: Timing Diagram for circuit element “X”

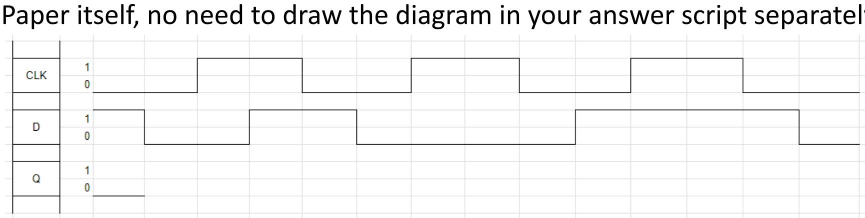
Design a negative-edge triggered D-flip flop using circuit element "X".</td><td>7</td></tr>
<tr><td></td><td>(b)</td><td>Complete the timing diagram in <u>figure 2</u> for the designed negative-edge triggered D-flip flop where D is the input, CLK is the clock signal and Q is the output to be drawn. (Use a pencil to draw the signal and draw on the Question Paper itself, no need to draw the diagram in your answer script separately)


Figure 2: Timing Diagram of Negative-edge triggered D flip-flop to be completed</td><td>8</td></tr>
</table>

4. (CO2)

Consider a sequential circuit with two inputs (***u1*** & ***u2***) and a single output, ***z***.
CLEO is a prestigious conference where researchers from all over the world submit their research articles. These articles are handled by an **Editor** and a **Reviewer**. They review each article *independently,* and either approve a submitted article (denoted by 1) or reject it (denoted by 0). The decisions of the **Editor** are given by a bit stream, ***u1***, whereas the decisions of the **Reviewer** are given by ***u2***. **If for three consecutive articles, both of their decisions are the same (i.e. *u1* = *u2* for three consecutive clock cycles), then they receive a consistency reward (*z* = 1) from the conference committee *after* a period.**
Here, the decisions of the Editor & the Reviewer for a number of submitted articles & their corresponding reward instances are given below:

u1	0	1	1	0	1	1	1	0	0	0	1
u2	1	1	1	0	0	0	1	0	0	1	1
z	0	0	0	0	1	0	0	0	0	1	0

Note:

- You **must** use Gray encoding to assign states.
- The overlapping of the input bits (***u1*** & ***u2***) is *not* allowed.

(a)	What type of FSM would you need for the given problem?	**1**
(b)	**Design** the state diagram for the corresponding FSM.	**4**
(c)	**Derive** the state-assigned table from your state diagram in (a).	**4**
(d)	**Evaluate** the <u>next state</u> & <u>output</u> logic expressions to implement the FSM & **calculate** the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation.)	**5 + 1**